Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Terra Dairy

Physical address of dairy:

8988 W Meadow DR	Winton	Merced	95388
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 05/10/1945

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0140-0030-0012-0000 0140-0120-0019-0000 0140-0150-0011-0000 0140-0160-0026-0000

B. OPERATORS

Terra, Edward Manuel Jr

Operator name: Terra, Edward Manuel Jr

Telephone no.: Landline: ; Cellular: (209) 769-5775

8988 W Meadow DR	Winton	CA	95388
Mailing Address Number and Street	City	State	Zip Code

Terra, Frank Joseph Sr

Operator name: Terra, Frank Joseph Sr

Telephone no.: Landline: ; Cellular: (209) 769-7853

8988 W Meadow DR	Winton	CA	95388
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Terra, Edward Manuel Jr

Legal owner name: Terra, Edward Manuel Jr

Telephone no.: Landline: ; Cellular: (209) 769-5775

8988 W Meadow DR	Winton	CA	95388
Mailing Address Number and Street	City	State	Zip Code

Terra, Frank Joseph Sr

Legal owner name:	Terra, Frank Joseph Sr	Telephone no.:		(209) 769-7853
			Landline	Cellular
8988 W Meadow DR		Winton	CA	95388
Mailing Address Number and Street		City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	0	0	0
Average number	0	0	0	0	0	0
Avg live weight (lbs)	0	0	0	0		

Predominant milk cow breed: Other

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

No fresh water sources entered.

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
First	43	43	0	none	0140-0030-0012-0000
Last	31	31	0	none	0140-0120-0019-0000
Middle	38	38	0	none	0140-0120-0019-0000
Pasture - North	38	38	0	none	0140-0150-0011-0000
Pasture - South	58	58	0	none	0140-0160-0026-0000
Totals for areas that were used for application					
Totals for areas that were not used for application	208	208	0		
Land application area totals	208	208	0		

B. CROPS AND HARVESTS

No application area fields entered.

NUTRIENT BUDGET

A. LAND APPLICATIONS

No application area crops entered.

B. NUTRIENT BUDGET

No application area crops entered.

NUTRIENT ANALYSES

A. MANURE ANALYSES

No manure analyses entered.

B. PROCESS WASTEWATER ANALYSES

No process wastewater analyses entered.

C. FRESH WATER ANALYSES

No irrigation water analyses entered.

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

No plant tissue analyses entered.

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

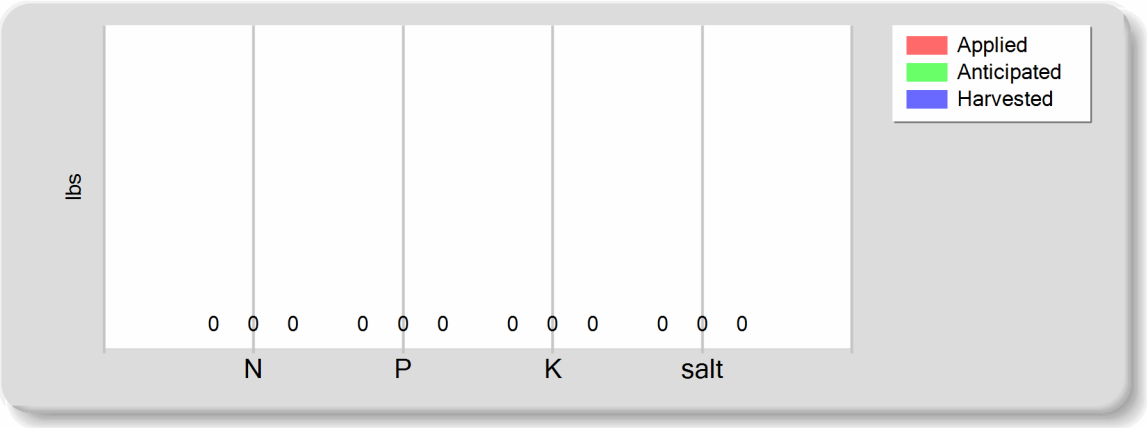
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	0.00	0.00	0.00	0.00
Total nutrients applied	0.00	0.00	0.00	0.00
Anticipated crop nutrient removal	0.00	0.00	0.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	0.00	0.00	0.00	0.00
Applied to removed ratio	0.00	0.00	0.00	0.00

B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL



C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE

Pounds of nitrogen applied

Existing soil nutrient content	0
Plowdown credit	0
Commercial fertilizer / Other	0
Dry manure	0
Process wastewater	0
Fresh water	0
Atmospheric deposition	0

lbs

Pounds of phosphorus applied

Existing soil nutrient content	0
Plowdown credit	0
Commercial fertilizer / Other	0
Dry manure	0
Process wastewater	0
Fresh water	0
Atmospheric deposition	0

lbs

Pounds of potassium applied

Existing soil nutrient content	0
Plowdown credit	0
Commercial fertilizer / Other	0
Dry manure	0
Process wastewater	0
Fresh water	0
Atmospheric deposition	0

lbs

Pounds of salt applied

Existing soil nutrient content	0
Plowdown credit	0
Commercial fertilizer / Other	0
Dry manure	0
Process wastewater	0
Fresh water	0
Atmospheric deposition	0

lbs

EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

1. Pasture - North & South had no nutrients applied or crops harvested by owner so no data was collected.

2. All Farming of Parcels 140-0120-019-0000 and 140-0030-0012-000 are leased to Kenneth And Johanna Fischer, Tierra Buena Ranch dba K. & J. Fischer 2008 Family Trust, in agreement the Lessee is responsible to obtain and report all samples, land applications and harvest events required by the California State Water Resources Control Board.

3. There were no milking operation at this facility during 2023

4. Included with report, two ground water monitoring lab results and chain of custody.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Joseph Terra, Sr	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

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CERTIFICATION

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Frank Terra Sr	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Joseph Terra, Sr	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/26/24	
DATE	DATE

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Eddie Terra Dairy
8988 W. Meadow Dr
Winton, CA 95388

Grower

Consultant:

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 12:41 PM
SUBMITTED BY: Frank
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 23.4

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well #1	T3362094A	Electrical Conductivity (EC)	EPA 120.1 and	0.645	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	413	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	97.4			0.230
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Eddie Terra Dairy
8988 W. Meadow Dr
Winton, CA 95388

Grower

Consultant:

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 12:41 PM
SUBMITTED BY: Frank
ANALYZED DATE: 1/3/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 23.4

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well #2	T3362094B	Electrical Conductivity (EC)	EPA 120.1 and	0.682	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	436	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	90.2	97.6 (T3362094B)	99.4 (T3362094B)	1.82
Electrical Conductivity (EC)	97.4			0.230
Nitrate Nitrogen (NO3-N)	101	92.2 (T3362014C)	91.3 (T3362014C)	1.00

he warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the ustomer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and onditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Eddie Terra Dairy Acct#5711
Mailing address: 8988 W. Meadow DR, Winton CA 95388
(not facility address)
Phone#: 209-769-7853

Authorized Copy Released To:

Sampled By: Frank Terra
Project ID: Dairy Compliance

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH3-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH3-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH3-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH3-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH3-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12/27/23	4pm	23.12.27 Domestic Well #1	Domestic well	23.4		✓				
2	12/28/23		23.12.28 Domestic Well #2	Domestic Well	23.4		✓				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	[illegible]	Terra Dairy				
2nd		DENELE ANALYTICAL, INC.	12-28-23 12:41			T332094